SAP Interface API Reference

VMM PO Governance — SAP Interface API Reference (v1.4)

For: SAP ABAP / workflow / Basis developers building the SAP side of the PO Governance check. **Contract:** *SAP API Specification v1.4* and `po-governance-sap-api.yaml` (OpenAPI 3.0) in `legacy/python-app/docs/`. This page lists every endpoint with its request and response as the tool answers them today. All examples below were captured from the running interface.

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1. Connection

Item	Staging (SAP QA)	Production
Base URL	`http://192.168.0.175:8178/api/sap/v1`	`https://<vm-dns-name>:8443/api/sap/v1` (company certificate)
SM59 destination	Type G, e.g. `ZPOGOV_API_TEST` — host `192.168.0.175`, port `8178`, path prefix `/api/sap/v1`, SSL off	Type G `ZPOGOV_API` — port `8443`, SSL active (import the company CA chain in STRUST)
Technical user	`sap_ecq` (one per SAP system)	`sap_ecp`
Connection test	`GET /ping` → `200`	same

Authentication

- **HTTP Basic authentication** with a technical user (SM59 → *Logon & Security* → *Basic Authentication*). The tool's administrator creates the user and gives you the password, generated with 32 characters, through a secure channel.
- Wrong or missing credentials → `401 UNAUTHORIZED`, with header `WWW-Authenticate: Basic`.
- After 5 wrong passwords in a row the user is locked for 15 minutes → `401 USER_LOCKED`.
- An optional IP allow-list can be switched on by the tool's administrator. A caller outside the list → `401 IP_NOT_ALLOWED`.

Rules for every call

- UTF-8 JSON, `Content-Type: application/json`. Field names are **snake_case**, exactly as written here.
- Timestamps are ISO 8601 with offset, e.g. `2026-10-22T14:24:58+05:30`. Dates are `YYYY-MM-DD`.
- Quantities are JSON numbers in the base unit (normally `PC`). Amounts are in document currency.
- Article numbers (`article`, `generic_article`) are sent in **SAP internal format with leading zeros**; the tool removes them.
- **OTB is never sent and never checked by the tool**; it stays in SAP.
- Every answer carries an `X-Trace-Id` header (e.g. `T-a1490d`). Quote it when you report a problem.

Error body (every non-200 answer)

```
{ "error_code": "VALIDATION_FAILED", "message": "purchase_order.items[0].order_qty must be > 0", "trace_id": "T-a19102" }
```

error_code	HTTP	Meaning	What SAP does
`UNAUTHORIZED`	401	Missing / wrong user or password	Fix SM59 credentials
`USER_LOCKED`	401	Too many wrong passwords, or user disabled	Ask the tool admin to unlock
`IP_NOT_ALLOWED`	401	Caller IP not in the allow-list (only when switched on)	Ask the tool admin to add the IP
`INVALID_JSON`	400	Body is not valid JSON	Fix the program; do not resend unchanged
`VALIDATION_FAILED`	400	Mandatory field missing, wrong value, quantity ≤ 0, bad date (`message` names the field)	Fix the program; do not resend unchanged
`REQUEST_ID_CONFLICT`	409	`/decisions`: same `request_id` already used with a different body	Generate a new `request_id`
`CANNOT_DECIDE`	422	`/decisions`: an item cannot be decided (e.g. article unknown to the tool)	Treat as **NOT_CHECKED**, continue as today
`BATCH_TOO_LARGE`	413	`/events`: more than 500 events or 5 MB	Split the batch and resend
`DECISION_TIMEOUT`, `DATA_NOT_LOADED`, `DATABASE_UNAVAILABLE`, `SAP_INTERFACE_DISABLED`, `INTERNAL_ERROR`	503	Tool temporarily unable to answer	`/decisions`: **NOT_CHECKED**, continue as today. `/events`: retry

2. Endpoints at a glance

#	Method & path	Called by	Purpose
1	`GET /ping`	SM59 connection test; health job every minute	Connection and credential check
2	`GET /reason-codes`	SAP approver screen	Exception reasons for the dropdown
3	`POST /decisions`	Release workflow background step "PO Governance check" (after OTB + depth check passed)	**Governance verdict per PO item** (synchronous, SAP waits max 3 s)
4	`POST /events`	bgRFC queue from BAdIs (after commit)	PO, goods receipt / return and governance-outcome events (asynchronous)

3. `GET /ping`

Connection test. If 3 health checks fail in a row, SAP switches the governance step off and runs the previous process.

Request: no body; Basic authentication only.

Response `200`:

```
{
  "status": "ok",
  "server_time": "2026-10-22T14:24:58+05:30",
  "api_version": "1.0.0",
  "environment": "TEST",
  "source_system_recognised": "ECQ"
}
```

Field	Type	Meaning
`status`	string	Always `ok` when the answer is 200
`server_time`	date-time	Tool server time
`api_version`	string	Interface version
`environment`	`TEST` | `PRODUCTION`	Which copy of the tool answered
`source_system_recognised`	string	SAP system ID registered for the technical user (`ECQ`, `ECP`) — check it matches `SY-SYSID`

Other answers: `401` (credentials), `503` (tool down).

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4. `GET /reason-codes`

Exception reasons the approver can choose when asking for an exception (send the chosen `code` back in a `GOV_OUTCOME` event). Maintained by Planning in the tool; read it when the screen opens or cache it daily.

Request: no body.

Response `200`:

```
{
  "version": "2026-10-22",
  "reason_codes": [
    { "code": "NEWL", "text": "New launch - no sales history", "active": true },
    { "code": "FEST", "text": "Festive / event build-up", "active": true },
    { "code": "VMOQ", "text": "Vendor minimum order quantity", "active": true },
    { "code": "PRCA", "text": "Price advantage / one-time buy", "active": true },
    { "code": "REPL", "text": "Store replenishment gap", "active": true },
    { "code": "OTHR", "text": "Other (note mandatory)", "active": true }
  ]
}
```

Field	Type	Meaning
`version`	string	Date of the last change — refresh your cache when it changes
`reason_codes[].code`	string, max 4	Code to show / store
`reason_codes[].text`	string, max 60	Text to show
`reason_codes[].active`	boolean	`false` = do not offer any more

The list above is the proposal from the specification and is **still to be confirmed by the business**.

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5. `POST /decisions` — the verdict API

Called by the workflow background step **after SAP's OTB check and depth plan check have passed**. SAP sends the PO; the tool answers with a verdict per item.

Timing and fallback contract

- The tool answers in **~20–60 ms** (measured; 50 parallel POs all under 1 s). SAP waits **at most 3 seconds**.
- On timeout, network error or **any non-200 answer**, SAP marks the PO **`NOT_CHECKED`** and continues the workflow exactly as today. **The tool can never stop SAP.**
- **Mode:** send `SHADOW` for now. The tool decides and logs, but SAP must **not** route on the verdict. `LIVE` = SAP routes on `route_to`. Control it in SAP, e.g. switch `ZPOGOV_ACTIVE` per division.

- **Idempotency:** `request_id` identifies one check (e.g. work-item ID + PO + version). Repeating the same `request_id` with the same body returns the **same answer** (same `decision_id`). The same `request_id` with a different body → `409 REQUEST_ID_CONFLICT`.

Request


```
{
  "request_id": "WI000004711-4500099001-V1",
  "mode": "SHADOW",
  "source_system": "ECQ",
  "client": "300",
  "otb_check": "PASSED",
  "depth_check": "PASSED",
  "previous_decision_id": null,
  "requested_at": "2026-10-21T11:50:00+05:30",
  "purchase_order": {
    "po_number": "4500099001",
    "doc_type": "NB",
    "vendor": "0000201345",
    "purch_org": "1000",
    "purch_group": "B07",
    "buyer_user": "BUYER07",
    "buyer_email": "ravi.k@vishalwholesale.co.in",
    "division": "MENS",
    "currency": "INR",
    "items": [
      {
        "po_item": 10,
        "article": "000000001101000101",
        "generic_article": "00000000001101000101",
        "colour": "WHT",
        "merch_category": "MC1021",
        "season": "AW26",
        "site": "DC01",
        "order_qty": 50000,
        "uom": "PC",
        "net_value": 418800.00,
        "delivery_date": "2026-11-15"
      }
    ]
  }
}
```


Field	Type	Req.	SAP source / meaning
`request_id`	string, max 80	**yes**	Unique per check, e.g. work-item ID + PO + version
`mode`	`SHADOW` | `LIVE`	**yes**	See above
`source_system`	string	**yes**	`SY-SYSID`
`client`	string	no	`SY-MANDT`
`otb_check`	`PASSED`	**yes**	Assertion that OTB passed (the tool is only called after it)
`depth_check`	`PASSED` | `BYPASSED`	**yes**	`BYPASSED` only for a quantity reduced under a previous REDUCE decision
`previous_decision_id`	string | null	no	On resubmission after a REDUCE: the `decision_id` it was reduced under
`requested_at`	date-time	no	When SAP sent the check
`purchase_order.po_number`	string	**yes**	`EKKO-EBELN`
`purchase_order.doc_type`	string	no	`EKKO-BSART`
`purchase_order.vendor`	string	no	`EKKO-LIFNR`
`purchase_order.purch_org`	string	no	`EKKO-EKORG`

Field	Type	Req.	SAP source / meaning
`purchase_order.purch_group`	string	no	`EKKO-EKGRP`
`purchase_order.buyer_user`	string	**yes**	Buyer's SAP user
`purchase_order.buyer_email`	string | null	no	Buyer's e-mail
`purchase_order.division`	string	no	Division (per-division rules and routing)
`purchase_order.currency`	string	no	`EKKO-WAERS`
`items[]`	array, min 1	**yes**	One entry per PO item
`items[].po_item`	integer	**yes**	`EKPO-EBELP`
`items[].article`	string	**yes**	`EKPO-MATNR` (variant / single article), leading zeros kept
`items[].generic_article`	string | null	**needed**	`MARA-SATNR` (generic article). The tool decides on the generic article; if empty it uses `article`
`items[].colour`	string | null	**needed**	Colour characteristic of the variant: Colour Master **code** preferred (description also accepted)
`items[].merch_category`	string	no	`EKPO-MATKL`
`items[].season`	string | null	no	Season
`items[].site`	string | null	no	`EKPO-WERKS` (informational; the tool works at company level)
`items[].order_qty`	number, > 0	**yes**	`EKPO-MENGE` in base unit
`items[].uom`	string	**yes**	`EKPO-MEINS`
`items[].net_value`	number	no	Item net value
`items[].delivery_date`	date	**needed**	First schedule line `EKET-EINDT` — the tool uses its month

"needed" means the specification allows the field to be empty, but the tool cannot decide an item without it. It then answers `422 CANNOT_DECIDE` (see below), so please always send these fields.

How the tool reads an item:

- **Article:** the generic article with its leading zeros removed (`000000001101000101` → `1101000101`), looked up in the tool's Article Master / Universe Master.
- **Colour:** matched by Colour Master code first, then by description.
- **Month:** the delivery month is taken from `delivery_date`. A date in the past counts as the current month.
- **Size variants:** items of one PO with the same generic article, colour and delivery month are decided **together** on their summed quantity. A REDUCE `max_qty` is split over them in proportion to their quantities.

Response `200`


```
{
  "decision_id": "D-20261022-000002",
  "request_id": "WI-REDUCE-0001",
  "mode": "SHADOW",
  "overall_verdict": "REDUCE",
  "verdict_code": "R",
  "route_to": "BUYER",
  "bm_head_required": false,
  "summary_text": "Item 10: reduce to 15,928 - Stock + pending + this PO cover 265 days, above the 7
9-day norm; max 15,928 pcs",
  "rules_version": "R-P1",
  "data_delayed": false,
  "data_as_of": { "bw_snapshot": "2026-10-22T14:24:36+05:30", "events_lag_seconds": 0 },
  "decided_at": "2026-10-22T14:24:58+05:30",
  "items": [
    {
      "po_item": 10,
      "verdict": "REDUCE",
      "verdict_code": "R",
      "max_qty": 15928,
      "reason_code": "COVER_ABOVE_NORM",
      "reason_text": "Stock + pending + this PO cover 265 days, above the 79-day norm; max 15,928 pc
s",
      "rule_id": "COVER-01",
      "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703, "headroom_qty": 15928,
"cover_days": 265 }
    }
  ]
}
```


Field	Type	Meaning / what SAP does with it
`decision_id`	string	e.g. `D-20261022-000002`. **Store it on the PO / workflow container** for audit and send it back in `GOV_OUTCOME` events
`request_id`	string	Echo of the request
`mode`	`SHADOW` | `LIVE`	Echo of the request
`overall_verdict`	`ALLOW` | `REDUCE` | `WATCH` | `HOLD`	Most severe item verdict (HOLD > WATCH > REDUCE > ALLOW)
`verdict_code`	`A` | `R` | `W` | `H`	One-character code for SAP domain values
`route_to`	`NONE` | `BUYER` | `CATHEAD` | `BMHEAD`	Where the workflow goes in LIVE mode (ignored in SHADOW). `NONE` = continue normal approvals
`bm_head_required`	boolean	Escalation to the B&M head (business rule pending — currently always `false`)
`summary_text`	string, max 255	Short text for the work item / PO header note
`rules_version`	string	Version of the planner parameters used
`data_delayed`	boolean	`true` = the tool's event feed is behind the agreed limit → **SAP continues the PO as today (NOT_CHECKED)**
`data_as_of.bw_snapshot`	date-time	Time of the stock data the decision used
`data_as_of.events_lag_seconds`	integer	How far behind the event feed is
`decided_at`	date-time	When the tool decided
`items[].po_item`	integer	PO item
`items[].verdict` / `verdict_code`	as above	Item verdict

Field	Type	Meaning / what SAP does with it
`items[].max_qty`	number | null	**Maximum allowed quantity when the verdict is REDUCE**; `null` otherwise
`items[].reason_code`	string	Machine-readable reason (table below)
`items[].reason_text`	string, max 132	Text for the buyer / approver
`items[].rule_id`	string	Rule that produced the verdict
`items[].metrics`	object	Figures behind the verdict. Single Concept: `plan_qty`, `stock_qty`, `open_po_qty`, `headroom_qty`, `cover_days`. Multi Concept: `sell_through_pct`. Fields that do not apply are omitted

`cover_days` is the cover from the first day of the month after delivery. It can be `0`: that means current stock plus pending POs is sold out before the delivery window starts.

Verdicts — what each one means for SAP

Verdict	Code	`route_to`	SAP action (LIVE mode)
`ALLOW`	A	`NONE`	Continue normal approvals
`REDUCE`	R	`BUYER`	Buyer reduces the item to `max_qty` or less. Depth plan bypass: store `decision_id` + `max_qty` on the item; if the quantity is reduced to `max_qty` or less, resubmit with `depth_check = BYPASSED` and `previous_decision_id`
`WATCH`	W	`CATHEAD`	Category head approves or rejects (exception with reason code)
`HOLD`	H	`CATHEAD`	Do not release as is; category head decides. **A blocked article (slow seller) is sent as `HOLD` with `reason_code = BLOCK`**

Reason codes per verdict

Verdict	reason_code	rule_id	Meaning
ALLOW	`COVER_WITHIN_NORM`	COVER-00	Stock + pending + this PO within the cover norm
ALLOW	`NO_PLAN_DATA`	PLAN-00	No sales plan for the bucket — allowed
ALLOW	`PEER_Q1` / `PEER_Q2`	PEER-Q1 / PEER-Q2	Multi Concept: ranks in the top half of its peer group
ALLOW	`PEER_Q3`	PEER-Q3-OK	Q3 but at / near the peer average (border zone)
ALLOW	`BOBS_SIBLINGS_OK`	BOBS-01	New colour of a selling style; sibling colours sell well
ALLOW	`LOW_HISTORY`	WOF-00	Too little selling history to judge
ALLOW	`NO_PEER_POOL`	PEER-00	No comparable styles
ALLOW	`NEW_ARTICLE`	NEW-00	New article (not in the tool's history)
REDUCE	`COVER_ABOVE_NORM`	COVER-01	Cover above the norm — reduce to `max_qty`
WATCH	`MIX_BELOW_MIN`	MIX-01	Colour share below the minimum mix (broken / cut size likely)
WATCH	`PEER_Q3`	PEER-Q3-GAP	Q3, beyond the border zone
WATCH	`PEER_Q4`	PEER-Q4-WATCH	Q4 but at / near the peer average
WATCH	`BOBS_SIBLING_LOW`	BOBS-02	A sibling colour sells poorly
HOLD	`COVER_NO_HEADROOM`	COVER-02	Stock + pending already above the norm — no headroom (max 0)
HOLD	`BLOCK`	BLOCK-Q4	Q4, well below the peer average: confirmed slow seller
HOLD	`BLOCK`	BLOCK-ABS	Absolute block: sell-thru below 1 %/day after 4+ weeks on floor

The tool may later send `BLOCK` (code `B`) as its own verdict instead of `HOLD` + `BLOCK`, once SAP has added that domain value. This is switched in the tool's settings and will be agreed with you first. Until then, handle `reason_code = BLOCK` under `HOLD`.

More examples (captured)

ALLOW, with the colour sent as its description (`"colour": "wht desc"`):


```
{ "decision_id": "D-20261022-000001", "request_id": "WI-ALLOW-0001", "mode": "SHADOW",
  "overall_verdict": "ALLOW", "verdict_code": "A", "route_to": "NONE", "bm_head_required": false,
  "summary_text": "All 1 item(s) within the norms", "rules_version": "R-P1", "data_delayed": false,
  "data_as_of": { "bw_snapshot": "2026-10-22T14:24:36+05:30", "events_lag_seconds": 0 },
  "decided_at": "2026-10-22T14:24:58+05:30",
  "items": [ { "po_item": 10, "verdict": "ALLOW", "verdict_code": "A", "max_qty": null,
               "reason_code": "COVER_WITHIN_NORM", "reason_text": "Cover 0 days within the 79-day no
rm", "rule_id": "COVER-00",
               "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703, "headroom_qt
y": 15928, "cover_days": 0 } } ] }
```


HOLD (no headroom):


```
{ "...": "...", "overall_verdict": "HOLD", "verdict_code": "H", "route_to": "CATHEAD",
  "summary_text": "Item 10: hold - Stock + pending already above the 79-day cover norm; no headroom
(max 0)",
  "items": [ { "po_item": 10, "verdict": "HOLD", "verdict_code": "H", "max_qty": null,
               "reason_code": "COVER_NO_HEADROOM", "reason_text": "Stock + pending already above the
79-day cover norm; no headroom (max 0)",
               "rule_id": "COVER-02",
               "metrics": { "plan_qty": 7108, "stock_qty": 25604, "open_po_qty": 3698, "headroom_qt
y": 0, "cover_days": 101 } } ] }
```


WATCH (colour mix):


```
{ "...": "...", "mode": "LIVE", "overall_verdict": "WATCH", "verdict_code": "W", "route_to": "CATHEA
D",
  "summary_text": "Item 10: watch - Colour mix 0.0% below the 2% minimum of its MC-MRP",
  "items": [ { "po_item": 10, "verdict": "WATCH", "verdict_code": "W", "max_qty": null,
               "reason_code": "MIX_BELOW_MIN", "reason_text": "Colour mix 0.0% below the 2% minimum
of its MC-MRP",
               "rule_id": "MIX-01", "metrics": { "stock_qty": 874, "open_po_qty": 0 } } ] }
```


Block (sent as HOLD + BLOCK):


```
{ "...": "...", "overall_verdict": "HOLD", "verdict_code": "H", "route_to": "CATHEAD",
  "summary_text": "Item 10: hold - Ranks Q4, sell-thru 64.4% below the peer average: confirmed slow
seller",
  "items": [ { "po_item": 10, "verdict": "HOLD", "verdict_code": "H", "max_qty": null,
               "reason_code": "BLOCK", "reason_text": "Ranks Q4, sell-thru 64.4% below the peer aver
age: confirmed slow seller",
               "rule_id": "BLOCK-Q4", "metrics": { "sell_through_pct": 0.97 } } ] }
```


Several items: two size variants of one generic article and colour (decided together, `max_qty` split 9,557 + 6,371 = 15,928) plus a Multi Concept item:

```
{ "decision_id": "D-20261022-000006", "request_id": "WI-MULTI-0001", "mode": "SHADOW",
  "overall_verdict": "REDUCE", "verdict_code": "R", "route_to": "BUYER", "bm_head_required": false,
  "summary_text": "Item 10: reduce to 9,557 - …; Item 20: reduce to 6,371 - …",
  "items": [
    { "po_item": 10, "verdict": "REDUCE", "verdict_code": "R", "max_qty": 9557, "reason_code": "COVE
R_ABOVE_NORM",
      "reason_text": "Stock + pending + this PO cover 265 days, above the 79-day norm; max 15,928 pc
s for 2 items together (50,000 pcs ordered)",
      "rule_id": "COVER-01", "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703,
"headroom_qty": 15928, "cover_days": 265 } },
    { "po_item": 20, "verdict": "REDUCE", "verdict_code": "R", "max_qty": 6371, "reason_code": "COVE
R_ABOVE_NORM",
      "reason_text": "Stock + pending + this PO cover 265 days, above the 79-day norm; max 15,928 pc
s for 2 items together (50,000 pcs ordered)",
      "rule_id": "COVER-01", "metrics": { "plan_qty": 6303, "stock_qty": 2707, "open_po_qty": 2703,
"headroom_qty": 15928, "cover_days": 265 } },
    { "po_item": 30, "verdict": "ALLOW", "verdict_code": "A", "max_qty": null, "reason_code": "PEER_
Q1",
      "reason_text": "Ranks Q1 in its peer group (sell-thru 5.2%/day): healthy seller", "rule_id":
"PEER-Q1",
      "metrics": { "sell_through_pct": 5.2 } } ] }
```


Error answers of `/decisions` (captured)

Case	HTTP	Body
Article unknown to the tool (whole PO not decided)	422	`{"error_code":"CANNOT_DECIDE","message":"Item 10: Article 9999999999 is unknown to the tool (not in Article / Universe Master, F2 or BSR)","trace_id":"T-a1490d"}`
Same `request_id`, different body	409	`{"error_code":"REQUEST_ID_CONFLICT","message":"request_id WI-ALLOW-0001 was already used with a different body (decision D-20261022-000001) - generate a new request_id","trace_id":"T-a4d243"}`
`order_qty` = 0	400	`{"error_code":"VALIDATION_FAILED","message":"purchase_order.items[0].order_qty must be > 0","trace_id":"T-a19102"}`
Broken JSON	400	`{"error_code":"INVALID_JSON","message":"Invalid JSON: …","trace_id":"T-1e66c8"}`
No / wrong credentials	401	`{"error_code":"UNAUTHORIZED","message":"Missing or wrong credentials","trace_id":"T-809985"}`

If **any** item cannot be decided, the whole request is answered `422`. The message names the item(s), and SAP treats the PO as NOT_CHECKED. Every non-200 answer from `/decisions` = **NOT_CHECKED, continue as today**.

---

6. `POST /events`

Called from a **bgRFC unit after commit**. One call carries **1–500 events** in the order they occurred (ascending `seq`).

Delivery contract

- `200` = every event in the batch is safely stored (as accepted, duplicate or parked). **Remove the unit from the queue.**
- Any other status, or no answer within **10 seconds → retry the same batch**. Resending is always safe: events are de-duplicated by `msg_id`.
- An event that fails validation is **parked** on the tool side (listed in `parked`, alerted to IT) and **never blocks the queue**.
- **PO events carry the full current state of the PO** (header + all items), not only what changed.

- Scope: vendor POs and goods movements against vendor POs (receipts 101/102, returns 122/161). Stock transfers (STO) are not sent.

Request

```
{
  "source_system": "ECQ",
  "client": "300",
  "queue": "ZPOGOV_EVENTS",
  "sent_at": "2026-10-21T11:42:08+05:30",
  "events": [ { "...one or more events, see below...": "" } ]
}
```

Field	Type	Req.	Meaning
`source_system`	string	**yes**	`SY-SYSID`
`client`	string	**yes**	`SY-MANDT`
`queue`	string	no	bgRFC queue name, e.g. `ZPOGOV_EVENTS`
`sent_at`	date-time	no	When the unit was sent
`events`	array, 1–500	**yes**	The events

Every event:

Field	Type	Req.	Meaning
`msg_id`	string, max 80	**yes**	Globally unique, built by SAP (system + client + object + timestamp / counter). Used for de-duplication
`seq`	integer (int64)	**yes**	Ascending number per queue, assigned when the event is written
`event_type`	enum	**yes**	One of the 11 types below
`occurred_at`	date-time	**yes**	When the document was saved in SAP
`sap_user`	string	no	`SY-UNAME`
`purchase_order` / `goods_movement` / `governance_outcome`	object	**one of them**	Filled according to `event_type`

event_type	Payload
`PO_CREATED`, `PO_CHANGED`, `PO_RELEASED`, `PO_RELEASE_REVOKED`, `PO_CLOSED`, `PO_DELETED`	`purchase_order` (full current state)
`GR_POSTED`, `GR_REVERSED`, `RETURN_POSTED`, `RETURN_REVERSED`	`goods_movement` (one material-document line)
`GOV_OUTCOME`	`governance_outcome` (what happened after a verdict)

`purchase_order` (PO events)

```
{
  "msg_id": "ECQ300-PO-4500098765-20261021101502-01", "seq": 45098, "event_type": "PO_RELEASED",
  "occurred_at": "2026-10-21T10:15:02+05:30", "sap_user": "BUYER07",
  "purchase_order": {
    "po_number": "4500098765", "doc_type": "NB", "vendor": "0000201345", "purch_org": "1000", "purch
_group": "B07",
    "created_on": "2026-10-18", "created_by": "BUYER07", "release_status": "RELEASED", "release_indi
cator": "R",
    "currency": "INR", "deleted": false,
    "items": [
      { "po_item": 10, "article": "000000001101000101", "generic_article": "000000001101000101", "co
lour": "WHT",
        "merch_category": "1101", "site": "DC01", "order_qty": 600, "received_qty": 0, "returned_qt
y": 0, "open_qty": 600,
        "uom": "PC", "net_price": 349.00, "price_unit": 1, "delivery_date": "2026-11-10",
        "deletion_flag": false, "delivery_completed": false }
    ]
  }
}
```

Field	Req.	SAP source
`po_number`	**yes**	`EKKO-EBELN`
`doc_type`, `vendor`, `purch_org`, `purch_group`	no	`EKKO-BSART`, `LIFNR`, `EKORG`, `EKGRP`
`created_on`, `created_by`	no	`EKKO-AEDAT / BEDAT`, `ERNAM`
`release_status`	no	`NOT_RELEASED` | `IN_RELEASE` | `RELEASED` | `REFUSED` (from `FRGKE` / `FRGZU`)
`release_indicator`, `currency`	no	`EKKO-FRGKE`, `WAERS`
`deleted`	no	`true` if the whole PO is deleted / cancelled
`items[]`	**yes**	All items
`items[].po_item`, `article`, `order_qty`, `uom`	**yes**	`EKPO-EBELP`, `MATNR`, `MENGE`, `MEINS`
`items[].generic_article`, `colour`	no (send them)	`MARA-SATNR`, colour characteristic
`items[].merch_category`, `site`	no	`EKPO-MATKL`, `WERKS`
`items[].received_qty`, `returned_qty`, `open_qty`	no	From `EKBE`; `open_qty = order − received + returned`, 0 if completed / deleted
`items[].net_price`, `price_unit`	no	`EKPO-NETPR`, `PEINH`
`items[].delivery_date`	no	First schedule line `EKET-EINDT`
`items[].deletion_flag`, `delivery_completed`	no	`EKPO-LOEKZ`, `ELIKZ`

`goods_movement` (GR / return events)

```
{
  "msg_id": "ECQ300-GR-5000123456-2026-0001", "seq": 45121, "event_type": "GR_POSTED",
  "occurred_at": "2026-10-21T11:42:07+05:30", "sap_user": "DCUSER01",
  "goods_movement": {
    "material_doc": "5000123456", "doc_year": 2026, "doc_item": 1, "movement_type": "101", "debit_cr
edit": "S",
    "posting_date": "2026-10-21", "article": "000000001101000101", "generic_article": "0000000011010
00101",
    "colour": "WHT", "site": "DC01", "storage_location": "0001", "quantity": 240, "uom": "PC",
    "po_number": "4500098765", "po_item": 10, "reversal_of": null
  }
}
```

Reversal (`GR_REVERSED`, movement 102) and return (`RETURN_POSTED`, movement 122):

```
{ "msg_id": "ECQ300-GR-5000123457-2026-0001", "seq": 45130, "event_type": "GR_REVERSED", "occurred_a
t": "2026-10-21T11:50:00+05:30",
  "sap_user": "DCUSER01",
  "goods_movement": { "material_doc": "5000123457", "doc_year": 2026, "doc_item": 1, "movement_typ
e": "102", "debit_credit": "H",
                      "posting_date": "2026-10-21", "article": "000000001101000101", "quantity": 40,
"uom": "PC",
                      "po_number": "4500098765", "po_item": 10, "reversal_of": "5000123456" } }

{ "msg_id": "ECQ300-RT-5000123500-2026-0001", "seq": 45140, "event_type": "RETURN_POSTED", "occurred
_at": "2026-10-21T12:00:00+05:30",
  "goods_movement": { "material_doc": "5000123500", "doc_year": 2026, "doc_item": 1, "movement_typ
e": "122", "debit_credit": "H",
                      "article": "000000001101000101", "quantity": 10, "uom": "PC", "po_number": "45
00098765", "po_item": 10 } }
```

Field	Req.	SAP source
`material_doc`, `doc_year`, `doc_item`	**yes**	`MBLNR`, `MJAHR`, `ZEILE`
`movement_type`	**yes**	`BWART` (101, 102, 122, 161 …)
`article`, `quantity`, `uom`	**yes**	`MATNR`, `MENGE` (always positive, base unit), unit
`debit_credit`	no	`S` | `H` (`SHKZG`) — the direction
`posting_date`	no	`BUDAT`
`generic_article`, `colour`	no (send them)	Generic article, colour
`site`, `storage_location`	no	`WERKS`, `LGORT`
`po_number`, `po_item`	no	`EBELN`, `EBELP` if the movement refers to a PO
`reversal_of`	no	Original material document (`SMBLN`) for a reversal

`governance_outcome` (`GOV_OUTCOME`)

What happened after a verdict (buyer accepted, exception approved / rejected …). It feeds the audit trail.

```
{
  "msg_id": "ECQ300-OUT-D-20261021-000001-01", "seq": 45188, "event_type": "GOV_OUTCOME",
  "occurred_at": "2026-10-21T15:05:12+05:30", "sap_user": "CATHEAD03",
  "governance_outcome": {
    "decision_id": "D-20261021-000001", "po_number": "4500099001", "po_item": 10,
    "outcome": "EXCEPTION_APPROVED", "by_user": "CATHEAD03", "by_role": "CATHEAD",
    "at": "2026-10-21T15:05:12+05:30", "new_qty": null, "reason_code": "FEST", "note": "Diwali build
-up for DC01"
  }
}
```

Field	Req.	Meaning
`decision_id`	**yes**	From the `/decisions` answer
`po_number`	**yes**	PO
`po_item`	no	Item (null = whole PO)
`outcome`	**yes**	`BUYER_ACCEPTED_REDUCTION` | `EXCEPTION_REQUESTED` | `EXCEPTION_APPROVED` | `EXCEPTION_REJECTED` | `PO_RELEASED` | `PO_WITHDRAWN` | `NOT_CHECKED_TIMEOUT`
`by_user`	**yes**	SAP user
`by_role`	no	`BUYER` | `CATHEAD` | `BMHEAD` | `SYSTEM`
`at`	**yes**	date-time
`new_qty`	no	New quantity (e.g. after accepting a reduction)
`reason_code`	no	Code from `GET /reason-codes`
`note`	no	Max 500 characters (mandatory for `OTHR` by business rule)

Response `200`


```
{
  "received_at": "2026-10-22T14:24:59+05:30",
  "accepted": [
    "ECQ300-PO-4500098765-20261021101502-01",
    "ECQ300-GR-5000123456-2026-0001",
    "ECQ300-GR-5000123457-2026-0001",
    "ECQ300-RT-5000123500-2026-0001",
    "ECQ300-OUT-D-20261021-000001-01"
  ],
  "duplicates": [],
  "parked": [
    { "msg_id": "ECQ300-GR-5000123999-2026-0001", "error_code": "VALIDATION_FAILED", "message": "goo
ds_movement.article is required" },
    { "msg_id": "ECQ300-XX-1", "error_code": "VALIDATION_FAILED",
      "message": "event_type PO_SHIPPED is not one of PO_CREATED, PO_CHANGED, PO_RELEASED, PO_RELEAS
E_REVOKED, PO_CLOSED, PO_DELETED, GR_POSTED, GR_REVERSED, RETURN_POSTED, RETURN_REVERSED, GOV_OUTCOM
E" }
  ]
}
```


Field	Meaning
`received_at`	When the tool stored the batch
`accepted`	`msg_id`s stored now
`duplicates`	`msg_id`s already received earlier (not stored again) — sending the same batch twice gives all of them here
`parked`	Events that failed validation, with the reason; IT is alerted. **Still a 200 — do not resend these**

Other answers: `400` for a malformed envelope (e.g. `{"error_code":"VALIDATION_FAILED","message":"client is required",…}`), `401` for credentials, `413` for more than 500 events or 5 MB (split), `503` for a temporary problem (retry).

---

7. Status of the tool side

Item	Status
`ping`, `reason-codes`, `decisions`, `events`	**Available for testing** on staging
Decision logic	Same rules as the tool's New Request screen (Single Concept, Multi Concept, Watch List, Max Permissible)

Item	Status
Data used for decisions	Uploaded BW / SAP extracts (stock, pending PO, sales, plan, BSR) + the day's approved requests
Events → live open-PO / stock position	**In development**: events are received and stored now, and will feed the live position. Until then decisions use the uploaded files, so run **SHADOW** mode
Separate `BLOCK` verdict, B&M head limit (`bm_head_required`), final reason-code list	To be confirmed with the business
Multi Concept `max_qty` (REDUCE for Multi Concept items)	Rules pending from the business

Suggested test sequence for the SAP team:

1. SM59 connection test (`GET /ping`).
2. `GET /reason-codes`.
3. `POST /decisions` in SHADOW mode for a few real POs (ALLOW, REDUCE, WATCH, HOLD cases), then the same call again to check you get the same `decision_id`.
4. Timeout / NOT_CHECKED handling (e.g. a wrong password gives 401 → must continue as today).
5. `POST /events` for a PO save, a goods receipt and a GOV_OUTCOME, then the same batch again to see `duplicates`.

The tool's administrator sees every call — request, response, timing and trace ID — on 🔌 **SAP Interface** in the portal, so results can be checked together.